

Topic	Decision / Details
Object-oriented programming language (if not using OOP, then justify the use of the language)	Python 3.12 (object-oriented; supports classes, inheritance, method overriding, polymorphism, and dynamic binding)
GUI library (strongly encouraged)	PyQt5 (Qt 5 bindings for Python; cross-platform)
IDE (Integrated Development Environment)	Visual Studio Code with the Microsoft Python extension
xUnit framework (e.g., JUnit for Java)	pytest (with parametrize for data-driven tests)
Programming style guide (must read it carefully)	PEP 8 (https://peps.python.org/pep-0008/) enforced with the Black code formatter
Project hosting site	https://github.com/MChaudhary123/Intro-to-Software-Engineering
Other decisions if applicable	Version control: Git + GitHub; virtual environment via python -m venv; dependencies pinned in requirements.txt

Part 2 - Unit Testing

Made a quick calculator and tested it.

```

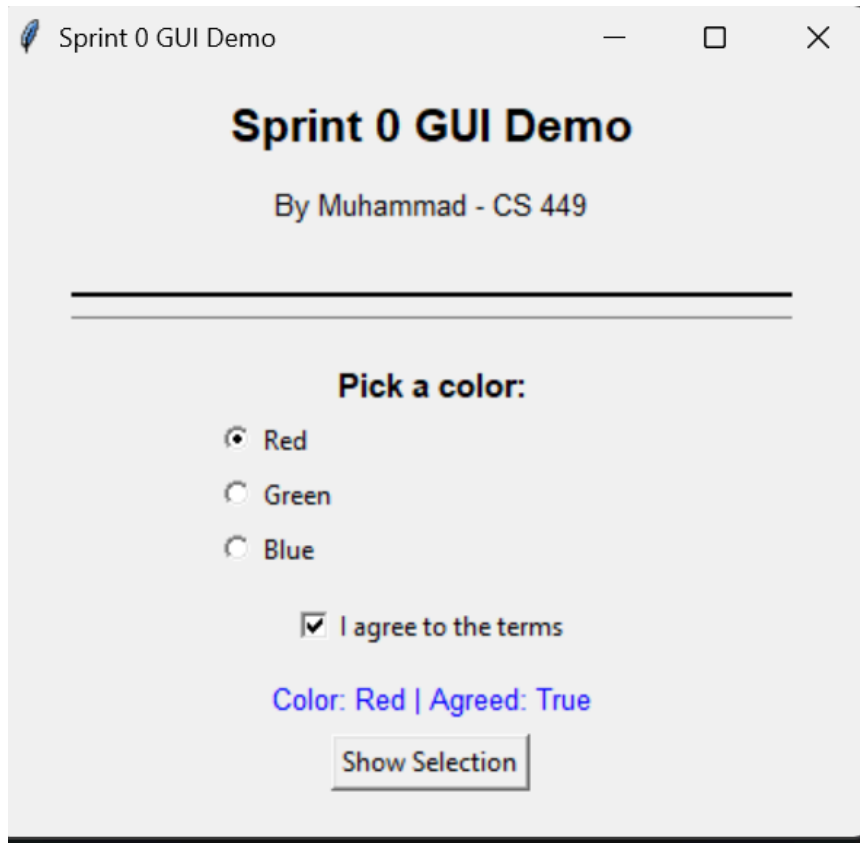
PS C:\Users\muham\OneDrive\2026 Fall semester 4th year\Foundations of SWE\Sprint 0> py -m pytest -v
===== test session starts =====
platform win32 -- Python 3.12.1, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\muham\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\muham\OneDrive\2026 Fall semester 4th year\Foundations of SWE\Sprint 0
collected 5 items

test_calculator.py::test_add PASSED [ 20%]
test_calculator.py::test_subtract PASSED [ 40%]
test_calculator.py::test_multiply PASSED [ 60%]
test_calculator.py::test_divide PASSED [ 80%]
test_calculator.py::test_divide_by_zero_raises PASSED [100%]

===== 5 passed in 0.16s =====
PS C:\Users\muham\OneDrive\2026 Fall semester 4th year\Foundations of SWE\Sprint 0>

```

Part 3 – GUI Programming



The screenshot shows a Java Swing window titled "Sprint 0 GUI Demo". The window has a light gray background and a standard title bar with a feather icon, a minus sign, a square icon, and a close button. The main content area contains the following elements:

- A title "Sprint 0 GUI Demo" in bold black font.
- A subtitle "By Muhammad - CS 449" in a smaller black font.
- A horizontal separator line consisting of two thin lines.
- A label "Pick a color:" in bold black font.
- Three radio buttons with labels "Red", "Green", and "Blue". The "Red" radio button is selected.
- A checkbox labeled "I agree to the terms" which is checked.
- A status text "Color: Red | Agreed: True" in blue font.
- A button labeled "Show Selection" with a thin black border.